

Shashi Bhushan Vijay

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SUMMARY

Results-driven Integrated M.Tech Computer Science student and patent holder specializing in machine learning pipelines, distributed systems, and backend engineering. Demonstrated ability to build production-grade systems achieving measurable outcomes—including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE & PROJECTS

quantFinance

Jan 2026 – Feb 2026

Convolve 4.0 – Pan-IIT AI/ML Hackathon

Repo

- Built a quant finance platform (Python, FastAPI, SQLite) to detect mispricing in AMZN option contracts via ensemble models (Black-Scholes, Heston) and generate trading signals.
- Implemented dynamic volatility forecasting (EWMA) and Greeks calculation for real-time risk management.
- Delivered a React frontend for signal monitoring and P&L tracking, providing end-to-end visibility from model to trade.

AutoML Assistant

Jan 2026

Project

Repo

- Developed a SaaS platform with a Streamlit frontend to automate dataset preprocessing, ML model training, and endpoint-based deployment without writing code.
- Engineered a FastAPI backend with LangChain + Groq (Llama 3.1) RAG to analyze dataset statistics and recommend optimal modeling strategies dynamically.
- Implemented a SHAP-powered explainability engine and live JSON/CSV batch inference endpoints different models.

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing

Dec 2024 – Dec

2025

VIT, Vellore, IN

Application No. 202541115892

- Engineered a FastAPI + MQTT backend processing real-time signals from ESP32/Raspberry Pi edge devices for reliable indoor localization where GPS fails.
- Developed a dynamic stacking ML ensemble (Scikit-learn, XGBoost) achieving **1.6 m accuracy ($R^2 = 0.978$)**, outperforming standard industry benchmarks.
- Deployed A* graph-based navigation for optimal path generation across unseen building layouts, delivering a complete patent-filed solution.

EDUCATION

Vellore Institute of Technology, Vellore

Expected April 2027

Integrated M.Tech in Computer Science Engineering

GPA: 7.0/10

SM Public School, Haridwar

2020–21

Junior College (Class XII)

91.2%

SM Public School, Haridwar

2018–19

High School (Class X)

87%

TECHNICAL SKILLS

Languages & Core: C++, Java, Python, JavaScript, SQL

Backend & Systems: FastAPI, REST APIs, Microservices, MQTT

Databases: PostgreSQL, SQLite | **Tools & Cloud:** Git, AWS, Streamlit

ML & AI: Scikit-learn, XGBoost, LSTM, SHAP, RAG, FAISS, LangChain, Groq, Quantitative Finance

LEADERSHIP & EXTRACURRICULARS

Sponsorship Coordinator, ISTE Chapter – VIT Vellore

- Secured **Rs. 1,00,000+ in corporate sponsorships** for flagship hackathon (Horizon) by pitching to 100+ founders and startups; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Built a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate | Designing Seamless Notification Service – View Certificate